

Customer Churn Data Analysis and Prediction

Task 1: Data Preparation

Dataset Overview

The dataset contains 7,043 customer records with 21 attributes, including demographic details (gender, senior citizen status, partner/dependents), account information (tenure, contract type, payment method), service usage (phone, internet, streaming, security add-ons), billing (monthly and total charges), and the target variable, Churn (whether the customer left).

customerID	gender	seniorCitizen	partner	dependents	tenure	phoneService	multipleLines	internetService	contractMonthToMonth	contractOneYear	contractTwoYear	deviceProtection	techSupport	streamingTV	streamingMovies	contract	paymentMethod	monthlyCharges	totalCharges	churn
7590-VHVEE	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	Yes	No	No	No	Month-to-month	Yes	29.85	29.85	No
5075-QHVEE	Male	0	No	No	34	Yes	No	DSL	Yes	No	No	No	No	No	No	One year	No	55.95	1989.5	No
3606-QYVBE	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	No	Month-to-month	Yes	53.85	108.15	Yes
7789-CFOCI	Male	0	No	No	45	No	No phone service	DSL	Yes	No	No	Yes	No	No	No	One year	No	42.3	1843.75	No
6274-HQDTU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	No	Month-to-month	Yes	70.7	151.65	Yes
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Yes	Month-to-month	Yes	99.65	820.5	Yes
1452-KZQW	Male	0	Yes	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	No	Month-to-month	Yes	89.1	1846.4	No
6713-KZQW	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No	No	No	No	No	Month-to-month	No	28.75	301.8	No
7862-FQDNE	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	No	Yes	Yes	Yes	Yes	Month-to-month	Yes	104.8	3068.05	Yes

Data Exploration

Initial exploration was done by checking column data types, counting unique categories in each text column, and scanning for missing values using `COUNTBLANK`. This revealed that all columns were complete except `TotalCharges`, which had 11 missing values.

Handling Missing Values

The 11 missing values in `TotalCharges` all corresponded to customers with `tenure = 0` — i.e., customers who had just signed up and hadn't been billed yet. These were filled with 0, since `TotalCharges` naturally starts at zero for a brand-new customer.

Categorical Variable Encoding

- `customerID` was dropped, since it's a unique identifier with no predictive value.
- Binary Yes/No columns (`gender`, `Partner`, `Dependents`, `PhoneService`, `PaperlessBilling`, `Churn`) were converted to 1/0 using `IF()` formulas.
- Multi-category columns (`MultipleLines`, `InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`, `Contract`, `PaymentMethod`) were one-hot encoded, creating a separate 0/1 column for each category (e.g., `Contract` became `Contract_MonthToMonth`, `Contract_OneYear`, `Contract_TwoYear`).
- This expanded the dataset from 21 columns to 41 fully numeric columns.

Lines_NoPhoneService	Lines_No	Lines_Yes	Internet_DSL	Internet_FiberOptic	Internet_No	Security_No	Security_Yes	Security_NoInternet
1	0	0	1	0	0	1	0	0
0	1	0	1	0	0	0	1	0
0	1	0	1	0	0	0	1	0
1	0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0	0
0	0	1	0	0	1	0	0	0
0	0	1	0	0	0	1	0	0
0	0	1	0	0	1	0	0	0
1	0	0	1	0	0	0	1	0
0	0	1	0	1	0	0	0	0
0	1	0	1	0	0	0	0	0
0	1	0	1	0	0	0	1	0
0	1	0	0	0	0	1	0	1
0	0	1	0	0	1	0	0	0
0	0	1	0	1	0	1	0	0

Train/Test Split

The cleaned dataset was randomly shuffled and split 80/20 into training and testing sets, resulting in 5,634 rows for training and 1,409 rows for testing. A fixed random seed (42) was used so the split is reproducible.

80% training split (5,634 rows), randomly sampled with seed=42.

gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	PaperlessBilling	MonthlyCharges	TotalCharges
0	0	1	0	1	0	1	24.8	24.8
1	0	0	0	41	1	1	25.25	996.45
0	0	1	1	52	1	0	19.35	1031.7
0	0	0	0	1	1	0	76.35	76.35
1	0	0	0	67	1	0	50.55	3260.1
0	1	1	0	68	1	1	89.6	6127.6
0	0	1	1	23	1	1	77.15	1759.4
1	0	1	1	72	1	0	72.1	5016.65
1	0	0	0	70	1	1	104	7250.15
1	0	0	0	1	1	0	19.4	19.4
1	0	0	0	1	1	0	20.45	20.45
4	0	4	4	24	4	4	84.05	2000.05

20% testing split (1,409 rows), randomly sampled with seed=42.

gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	PaperlessBilling	MonthlyCharges	TotalCharges
1	0	0	0	1	1	1	45.1	45.1
0	0	1	0	53	1	1	106.1	5769.75
0	0	1	0	14	1	0	105.95	1348.9
1	0	1	1	66	0	0	54.65	3632
1	0	1	0	68	1	1	100.3	6754.35
0	0	0	0	28	1	0	54.65	1517.5
0	0	1	1	3	0	0	29.7	91.7
0	0	0	0	24	1	1	94.6	2283.15
0	0	0	0	9	0	0	40.75	359.4
4	0	4	0	22	4	4	20.45	474.55

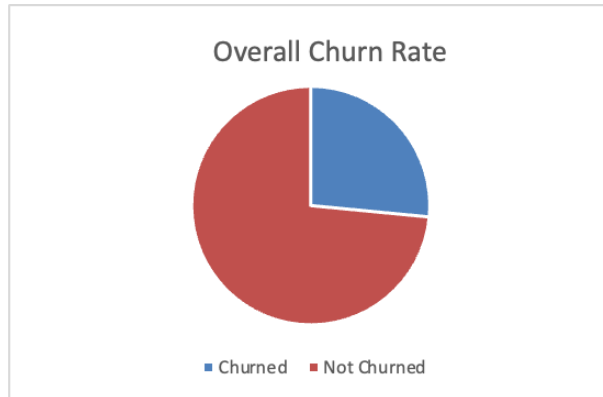
Summary of Task 1

After preparation, the dataset had no missing values, all categorical variables were converted to numeric form, and the data was split into training and testing sets ready for model building in the next task.

Task 2: Exploratory Data Analysis (EDA)

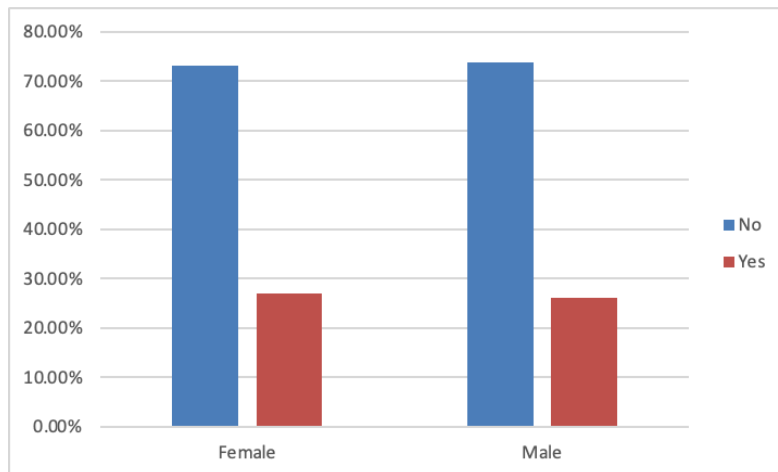
Overall Churn Rate

Out of 7,043 total customers, 1,869 churned (cancelled their service), giving an overall churn rate of **26.54%**. This means roughly 1 in 4 customers leave.



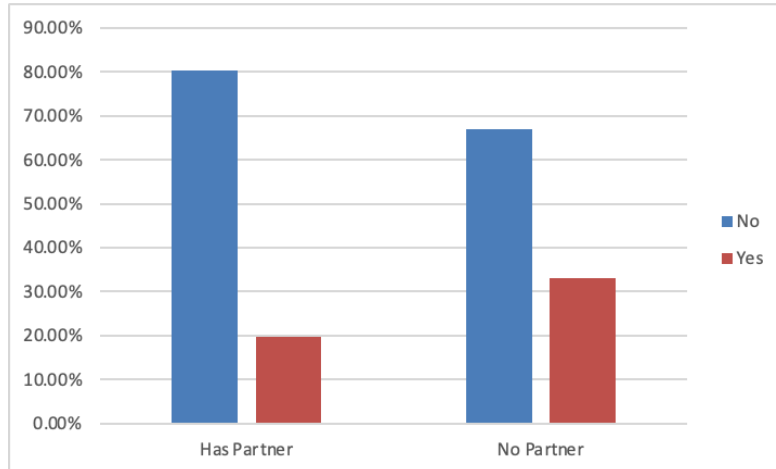
Churn by Gender

Churn rate was nearly identical between genders: 26.9% for Female customers and 26.2% for Male customers. This suggests gender has little to no relationship with churn.



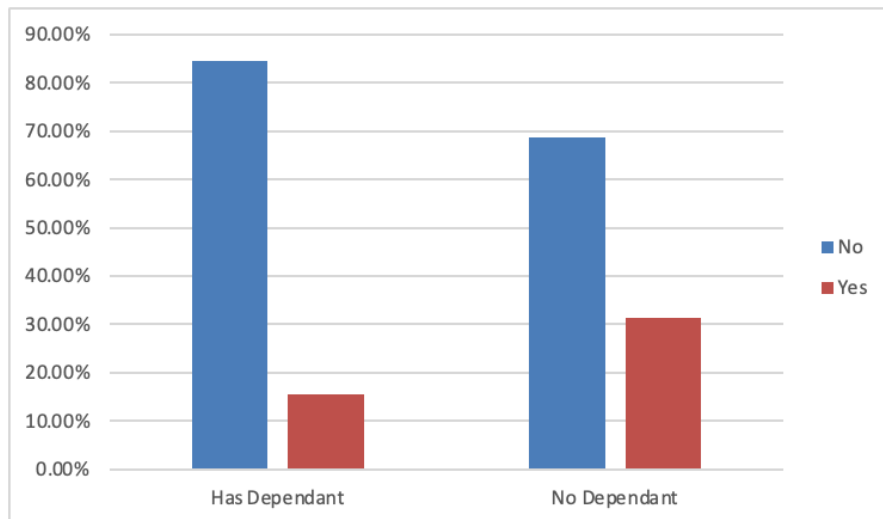
Churn by Partner Status

Customers with a partner churned significantly less (19.7%) than customers without a partner (33.0%). Having a partner appears to be a meaningful predictor of customer retention — partnered customers may be more settled/stable and less likely to switch providers.



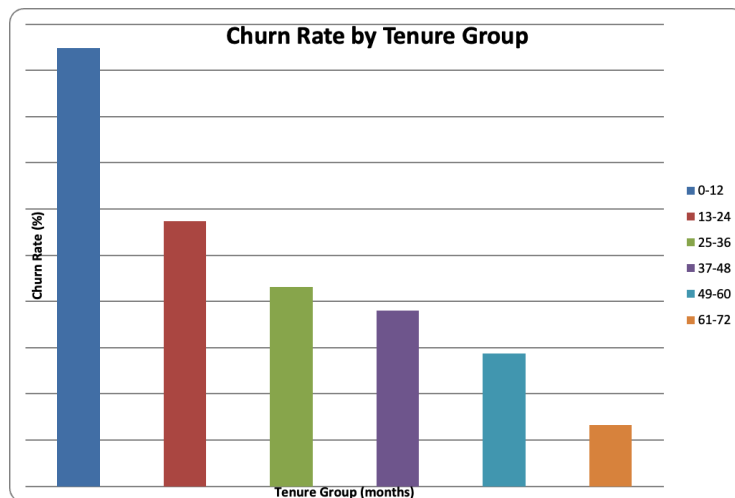
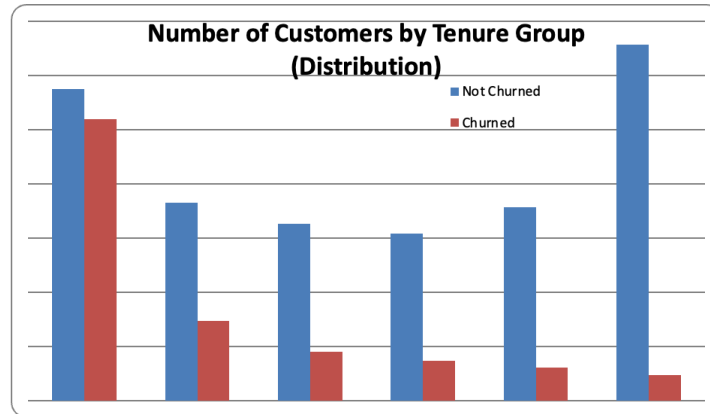
Churn by Dependent Status

Customers with dependent had 15.5% churn rate while without dependent had churn rate of 31.3%



Tenure Distribution and Its Relation to Churn

Churn rate drops sharply as tenure (length of time as a customer) increases: customers in their first year churn at about 47%, while customers with 5+ years of tenure churn at only 6.6%. This shows tenure is one of the strongest indicators of churn risk — newer customers are far more likely to leave, while long-term customers are much more loyal.

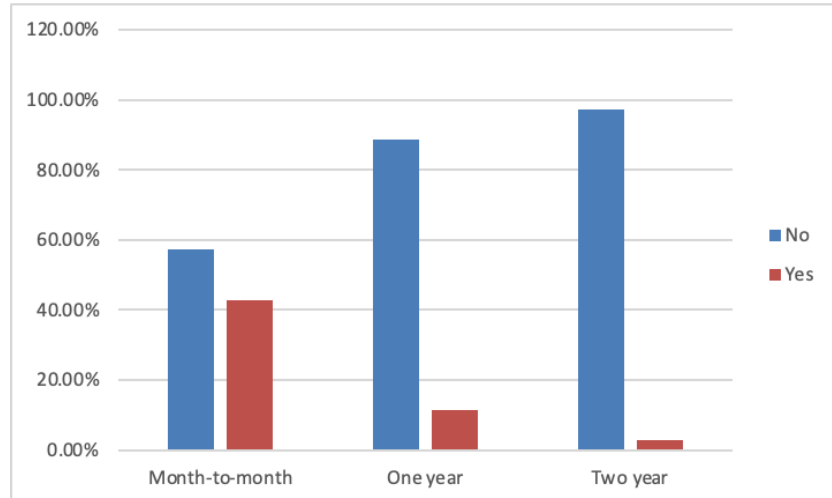


Churn by Contract Type

Contract type showed one of the strongest relationships with churn:

- Month-to-month contracts: 42.7% churn
- One-year contracts: 11.3% churn
- Two-year contracts: 2.8% churn

Customers on flexible, short-term contracts are far more likely to leave, while long-term contracts strongly encourage retention.



Churn by Payment Method

Customers paying by electronic check churned at a notably higher rate (45.3%) compared to automatic payment methods like bank transfer (16.7%) and credit card (15.2%). This may reflect that electronic check users are less "locked in" to automatic recurring billing, making it easier for them to leave.

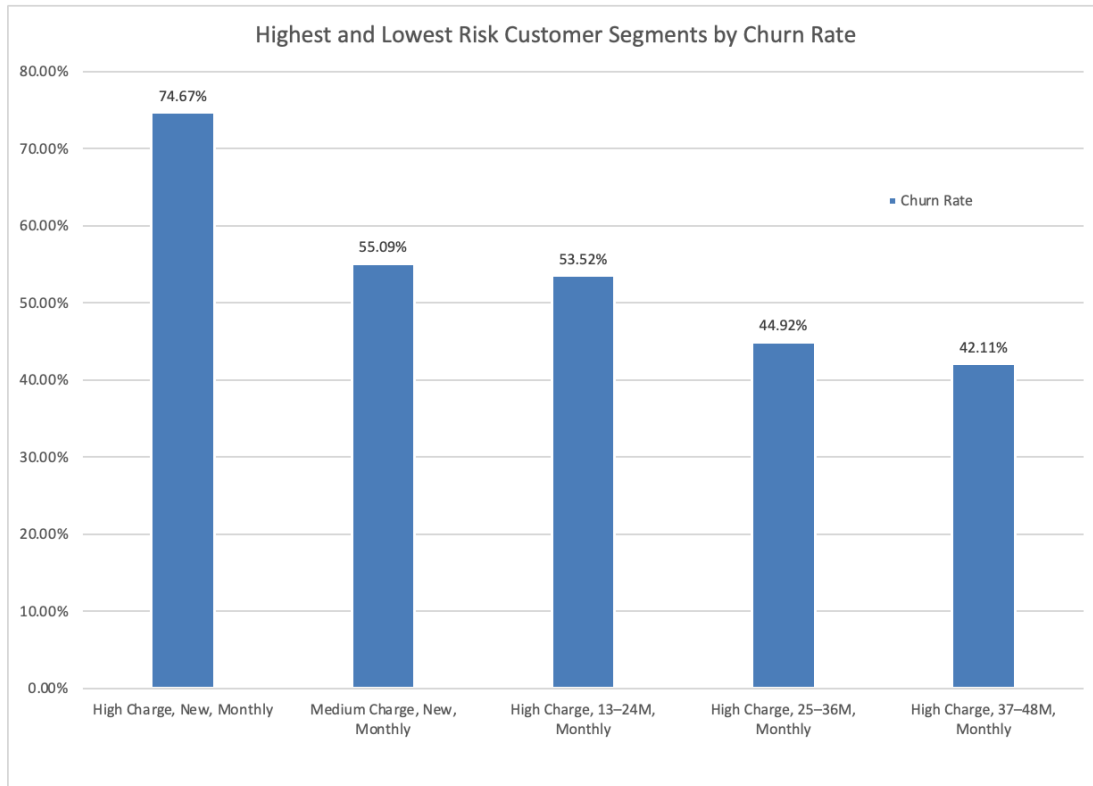
Summary

The EDA revealed that demographic factors like gender have little influence on churn, while behavioral and contractual factors — tenure, contract type, partner status, and payment method — are strong indicators of churn risk. Customers who are new, on month-to-month contracts, without a partner, and paying by electronic check represent the highest-risk group for churn. These findings will help guide feature selection and interpretation in the modeling task ahead.

Task 3: Customer Segmentation

Segmentation Approach

Customers were segmented based on three key factors: monthly charges, tenure, and contract type. Monthly charges were grouped into Low, Medium, and High categories, while tenure was divided into intervals (0–12 months, 13–24 months, etc.) to identify patterns among different customer groups.



Churn Analysis Within Segments

The segmentation analysis showed that the highest churn risk came from customers with **high charges, short tenure, and month-to-month contracts**. This segment had the highest churn rate at **74.67%**, indicating that newer high-value customers are more likely to leave. Other high-risk groups included medium-charge customers in their first year with month-to-month contracts (55.09%) and high-charge customers with 13–24 months tenure on month-to-month contracts (53.52%).

In contrast, customers with longer tenure and two-year contracts showed very low churn rates, suggesting that long-term contracts are strongly associated with customer retention.

High-Value Customers at Risk

High-charge customers on flexible month-to-month contracts were identified as priority retention targets. Offering loyalty incentives, contract benefits, or early engagement strategies could help reduce churn among these valuable customers.

Summary

Customer segmentation revealed that churn is not evenly distributed across customers. The combination of **high monthly charges, shorter tenure, and month-to-month contracts** represents the highest-risk segment, providing clear targets for retention efforts.

Task 4: Churn Prediction Model

Objective

Build machine learning models to predict customer churn and identify key drivers of customer attrition.

Methodology

Data Preparation

- Loaded cleaned dataset from Task 1 (7,043 customers, 43 numeric features)
- Target variable: Churn (0 = No, 1 = Yes)
- Train/Test Split: 80% (5,634) / 20% (1,409)
- Scaled numerical features: tenure, MonthlyCharges, TotalCharges

Models Trained

Model	Description
Logistic Regression	Baseline linear classifier
Decision Tree	Rule-based classifier
Random Forest	Ensemble of decision trees
Gradient Boosting	Sequential ensemble learning

Results

Model Performance Comparison

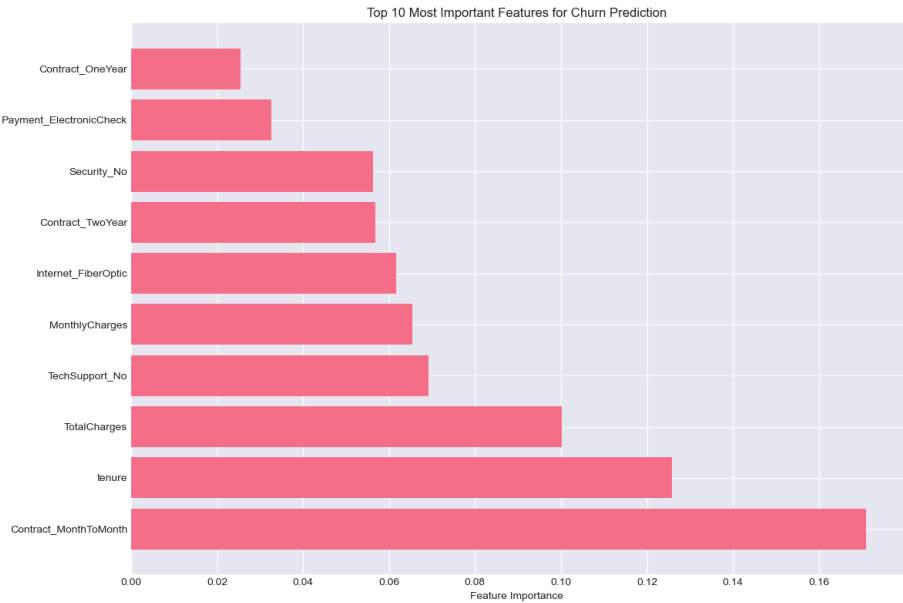
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	73.8%	50.4%	78.3%	61.4%	84.1%
Decision Tree	73.6%	50.2%	67.6%	57.6%	78.7%

Random Forest	76.2%	53.6%	77.8%	63.5%	84.3%
Gradient Boosting	79.1%	63.0%	51.3%	56.6%	83.9%

Best Model: Random Forest

Metric	Score	Interpretation
Accuracy	76.2%	Correctly predicts 3 out of 4 customers
Recall	77.8%	Catches 78% of actual churners
F1-Score	63.5%	Best balance of precision and recall
ROC-AUC	84.3%	Excellent predictive power

Feature Importance



Top 5 Churn Drivers

Rank	Feature	Importance	Business Meaning
1	Contract_MonthToMonth	17.1%	Month-to-month contracts drive churn
2	tenure	12.6%	New customers are high risk
3	TotalCharges	10.0%	Higher charges correlate with churn
4	TechSupport_No	6.9%	No tech support increases churn
5	MonthlyCharges	6.5%	Higher monthly bills = higher risk

Key Insight: Contract type + tenure account for ~30% of predictive power.

Summary

Best Model: Random Forest (F1: 63.5%, ROC-AUC: 84.3%)

Top Driver: Month-to-month contracts (17.1% importance)

Task 5: Model Evaluation & Interpretation

Model Performance

Random Forest was selected as the best model with:

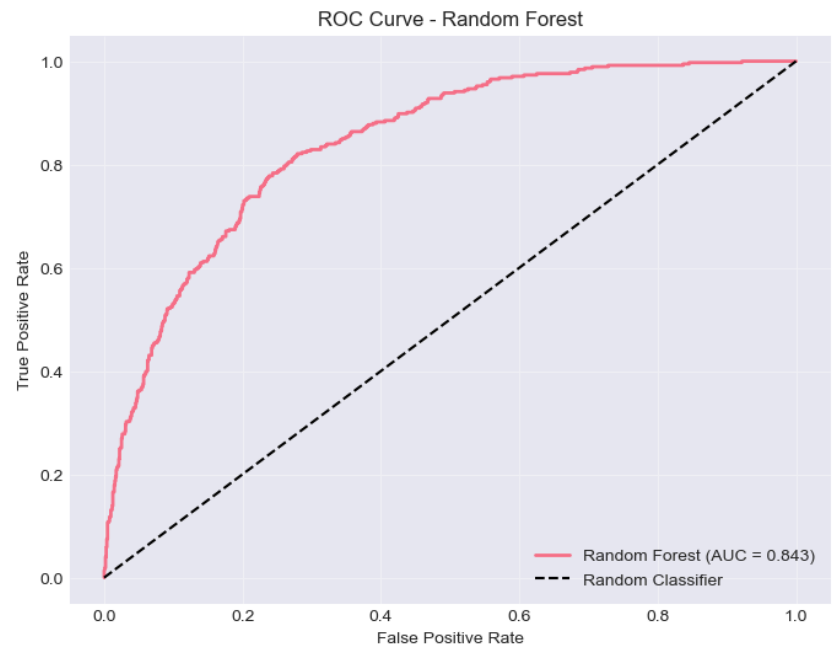
- Accuracy: 76.2%
- Recall: 77.8% (catches 3 out of 4 churners)
- ROC-AUC: 84.3% (excellent discrimination)

Confusion Matrix

	No	Yes
Actual No	850	100
Actual Yes	109	350

Metric	Value
True Negatives	850
False Positives	100
False Negatives	109
True Positives	350

ROC Curve Analysis



Random Forest ROC-AUC: 0.843

The ROC curve shows the model achieves excellent separation between churners and non-churners. An AUC of 0.843 means the model correctly ranks a randomly chosen churner higher than a non-churner 84.3% of the time.

Feature Importance Interpretation

Feature	Importance	Actionable Insight
Contract_MonthToMonth	17.1%	Converting to longer contracts should be top priority
tenure	12.6%	New customers need immediate engagement
TechSupport_No	6.9%	Tech support is a valuable retention tool

Model Validation

- Cross-Validation F1: 0.62 ± 0.02 (consistent performance)
- Conclusion: Model generalizes well to new data

Task 6: Business Recommendations

Priority Actions

Priority	Action	Expected Impact
1	Convert month-to-month to yearly contracts	Highest
2	New customer retention (< 12 months tenure)	High

3	Encourage automatic payment methods	Moderate
4	Bundle tech support as value-add	Moderate
5	Target high-risk segments (>70% probability)	High

Implementation Plan

1. Contract Conversion Program

- Target: Month-to-month customers
- Offer: 10-15% discount for 1-year commitment
- Expected: 30-40% churn reduction

Current Month-to-Month Churn: 42.7%

Current 1-Year Contract Churn: 11.3%

Potential Reduction: $42.7\% - 11.3\% = 31.4\%$

Estimated Impact: 30-40%

2. New Customer Retention

- Target: Tenure < 12 months
- Tactics: Welcome calls, check-ins at 30/60/90 days
- Expected: 25-35% churn reduction

3. Payment Method Incentives

- Target: Electronic check users
- Offer: \$5/month discount for auto-pay
- Expected: 10-15% churn reduction

4. Tech Support Bundles

- Target: High-risk customers
- Offer: 3-month free tech support trial
- Expected: 15-20% churn reduction

5. High-Risk Campaigns

- Target: >70% churn probability
- Tactics: Personalized retention offers
- Expected: 20-30% churn reduction

Revenue Impact

Average Monthly Charge from dataset: ~\$50 per customer

Annual Revenue: $\$50 \times 12 \text{ months} = \600 per customer

Churned Customers: $7,043 \times 26.54\% = 1,869$

Annual Revenue Per Customer: \$600

Revenue Lost: $1,869 \times \$600 = \$1,121,400$

Scenario	Churn Reduction	Annual Revenue Saved
Conservative	10%	$7,043 \times 26.54\% \times \$600 \times 0.10 =$ ~\$112,000
Moderate	20%	$7,043 \times 26.54\% \times \$600 \times 0.20 =$ ~\$224,000
Aggressive	30%	$7,043 \times 26.54\% \times \$600 \times 0.30 =$ ~\$337,000

Summary of Recommendations

Focus: Month-to-month customers and new customers

Action: Contract incentives + retention programs

Goal: Reduce churn by 20% (\$224,000+ annually)

Monitor: Track churn rate, contract conversions, and retention offer acceptance

Disclaimer

Based on the analysis, we recommend these initiatives. Actual results will depend on implementation quality and customer response. We suggest phased testing to measure real impact.